

Kantipur City College
Department of Computer Engineering
Semester IV/ Probability and Statistics
Lab Sheet No.7

Title: Expectations and Random Variables

Objectives: To find Expectations and Variance by using Excel.

Theory:

Mathematical Expectations:

If X is a discrete random variable which can assume any one of the values $x_1, x_2, \dots, x_n$ with respective probabilities $p(x_1), p(x_2), \dots, p(x_n)$, then the mathematical expectation of random variable X usually called the expected value of x is defined as

$$E(X) = x_1p(x_1) + x_2p(x_2) + \dots + x_np(x_n) \\ = \sum x p(x) \text{ where } \sum p(x) = 1$$

Rules of Expectation

- i) If a is any constant then $E(a) = a$
- ii) If a and b are constants then $E(ax \pm b) = a E(x) \pm b$
- iii) If X and Y are two r.v then $E[X+Y] = E(X) + E(Y)$
- iv) If X and Y are two independent r.v then $E(XY) = E(X) \cdot E(Y)$

Variance

Let X have pmf $p(x)$ and expected value μ . Then the variance of X , denoted by $V(X)$ or σ^2 is

$$V(X) = E[(X - \mu)^2] = \sum (X - \mu)^2 p(x)$$

The standard deviation (SD) of X is

$$\sigma_x = \sqrt{\sigma^2_x}$$

A shortcut formula for σ^2

$$\sigma^2 = V[X] = E(X^2) - [E(X)]^2 \\ = \sum x^2 \cdot p(x) - \mu^2$$

Rules of variance

- i. If a is any constant $V(a) = 0$

ii. If a and b are constants, then

$$V(aX + b) = a^2 V(X)$$

iii. If a and b are constants and X and y are two independent r.v, then

$$V(aX \pm bY) = a^2 V(X) + b^2 V(Y)$$

Lab work :

1. Find the expected value and its variance of the sum of number of points on faces of a dices when two dice are thrown.

Using Excel:

The values in row 3 indicate values in one dice and the values in column B gives the values in dice 2 and values in D4 to H9 gives the sum of outcomes in two dice.

	A	B	C	D	E	F	G	H
1								
2				Dice 1				
3			1	2	3	4	5	6
4		1	2	3	4	5	6	7
5		2	3	4	5	6	7	8
6	Dice 2	3	4	5	6	7	8	9
7		4	5	6	7	8	9	10
8		5	6	7	8	9	10	11
9		6	7	8	9	10	11	12

11	Let X be the random variable denoting the sum of points obtained from pair of dice, x can take the values 2,3,.....12				
12					
13		x	P(X=x)	xP(X=x)	x ² P(X=x)
14			=COUNTIF(\$C\$4:\$H\$9,B14)/COUNT(\$C\$4:\$H\$9)	=B14*C14	=B15^2*C15
15		2	0.02778	0.055555556	0.111111111
16		3	0.05556	0.166666667	0.5
17		4	0.08333	0.333333333	1.333333333
18		5	0.11111	0.555555556	2.777777778
19		6	0.13889	0.833333333	5
20		7	0.16667	1.166666667	8.166666667
21		8	0.13889	1.111111111	8.888888889
22		9	0.11111	1	9
23		10	0.08333	0.833333333	8.333333333
24		11	0.05556	0.611111111	6.722222222
25		12	0.02778	0.333333333	4
26		Total	1.00000	7	54.83333333
27			=SUM(C14:C24)	=SUM(D15:D25)	=SUM(E15:E25)
28					

2. The probabilities that a man fishing at Janakpur will catch 0, 1, 2,3,4,5 fishes are 0.10, 0.15, 0.30, 0.20, 0.15,0.10 respectively. Compute $E(4x+7)$ and Standard Deviation of $(4X+7)$.

Using Excel;

	A	B	C	D
1				
2				
3				
4	Table to calculate E(x) and V(X)			
5				
6	X	P(X)		
7	0	0.1		
8	1	0.15		
9	2	0.3		
10	3	0.2		
11	4	0.15		
12	5	0.1		
13				
14		Value	Formula	
15	E(X)	2.45	=SUMPRODUCT(A7:A12,B7:B12)	
16	E(X ²)	8.05	=SUMPRODUCT(A7:A12^2,B7:B12)	
17	V(X)	2.0475	=B16-(B15)^2	
18	E(4X+7)	16.8	=4*B15+7	
19	V(4X+7)	32.76	=4^2*B17	
20	S.D. of (4X+7)	5.723635	=SQRT(B19)	

3. Following information are given;

Project	Sales(Rs.'000)	Prob	Cost(Rs.'000)	Prob
1	12	0.2	6	0.3
2	15	0.3	12	0.2
3	20	0.6	8	0.4
4	28	0.4	15	0.1
5	30	0.5	20	0.7

Which Project would you select and why?

Using Excel;

	A	B	C	D	E	F	G	H
1								
2								
3		Project	Sales('000)	Prob	Cost('000)	Prob	Profit	Formula
4		1	12	0.2	6	0.3	0.6	=PRODUCT(C4:D4)-PRODUCT(E4:F4)
5		2	15	0.3	12	0.2	2.1	
6		3	20	0.6	8	0.4	8.8	
7		4	28	0.4	15	0.1	9.7	
8		5	30	0.5	20	0.7	1	
9								
10		The maximum expected profit is 9.7, which is obtained from Project 4. Hence we select Project 4.						

Assignments:

1. A random variable X has the following probability function.

X	-2	-1	1	1	2
P(X)	0.2	0.1	0.4	0.2	0.1

Compute $E(X)$, $E(4X+5)$, $V(X)$, $V(4X+5)$. Also compute standard deviation of X and $(4X+5)$.

2. An investor to asset in one of the following projects whose estimated profit/loss is given below

Project	Profit(Rs.'000)	Probability	Loss(Rs.'000)	Probability
1	35	0.5	15	0.5
2	48	0.3	25	0.7
3	45	0.6	20	0.4